Strategy Theory Notes

The mechanics, and the theory behind them, of every model in Model Evaluation Summary.txt

Covers Sigma7p, Sigma8, SigmaB10, Sigma10X, Sigma10FB, Sigma16, and Sigma17: cointegrated pair trading, factor-residualized lead-lag forecasting, tiered consensus voting, structural-break detection, and the five ways this project has used the ALGO slot to control market exposure.

Contents

How to Read This Document 3

The Strategy Building Blocks 4

A. Seed Pairs: Cointegrated Pair Trading . . . 4

B. The Two-Factor Lead-Lag Matrix . . . 6

C. The Gray / Survivor Tier . . . 7

D. The Birth-Catcher . . . 8

E. Tail-Regime Diagnostics and Hysteresis . . . 9

The Exposure-Control Mechanics 11

1. Pure Beta Hedge (Sigma7p, Sigma8) . . . 11

2. Directional Reversal Bet (SigmaB10, Sigma16, half of Sigma17) . . . 12

3. Anti-Hedge / Leverage (Sigma10X) . . . 13

4. Flat / Untraded (Sigma10FB) . . . 14

5. Relative Reversal (half of Sigma17, weight 0.5) . . . 15

Cross-Reference: Models to Mechanics 17

Appendix: The Scoring Function 18

How to Read This Document

Companion to Model Evaluation Summary.txt and Model Performance Evaluation.txt.

Every model in the summary is built from the same small set of Lego bricks. Part One walks through the five architectural building blocks shared by the whole Sigma family (what finds a tradeable relationship, and how much to trust it). Part Two walks through the five different things this project has tried putting in the ALGO slot, the one instrument with a 10× larger position limit and 5× lower commission, and therefore the one decision that swings total book performance the most.

Each block below states the mechanism as implemented in code, then the established theory it corresponds to, then (where this project measured it directly) what actually happened when the assumption was tested against real data. Boxed callouts marked KEY IDEA are the one-sentence takeaway if you read nothing else in that section.

Symbol	Meaning
`S_t`	spread or signal level at time t
`ϕ (phi)`	AR(1) autoregressive coefficient of a spread
`κ (kappa)`	OU mean-reversion speed; larger = faster reversion
`z_t`	signal standardized by its own rolling mean/vol
`β (beta)`	regression sensitivity of a name to the ALGO factor
`L`	the 50×50 dense one-lag lead-lag forecast matrix
`IC / BR`	information coefficient / breadth (Grinold's law)
`μ, σ`	mean and standard deviation of daily PnL
`SR, Score`	annualized Sharpe and the competition's scoring function

Part One

The Strategy Building Blocks

Five components, layered on top of each other in roughly the order they were added to the Sigma family. Every model in the summary is some subset of these five, plus one exposure mechanic from Part Two.

A. Seed Pairs: Cointegrated Pair Trading

Present in every model. Two instruments are cointegrated if some linear combination of their log-prices is stationary, even though each price series individually is a non-stationary random walk. If log P_i and log P_j are each a random walk (variance grows without bound), but the spread below reverts to a fixed mean, then i and j share a cointegrating vector (1, -β). This is the classic Engle–Granger framework (1987): estimate β by regression, then test the residual spread for stationarity.

$$S_t = \log P_{i,t} - \beta \log P_{j,t}$$

cointegrating spread

The code's `ar1_t` function fits an AR(1) model to the spread and converts the fitted coefficient into a t-statistic against the null ϕ=1 (a unit root, i.e. no reversion at all): a simplified Dickey–Fuller test.

$$S_t = \phi\, S_{t-1} + \varepsilon_t \qquad t_\phi = \frac{\hat{\phi} - 1}{SE(\hat{\phi})}$$

AR(1) unit-root test

The underlying process is best thought of as an Ornstein–Uhlenbeck (OU) process, the continuous-time model of mean reversion used throughout quantitative finance:

$$dS_t = -\kappa(S_t - \mu)\,dt + \sigma\, dW_t$$

Ornstein-Uhlenbeck spread dynamics

κ (kappa) sets the speed of reversion, and any deviation decays with a half-life of ln 2 / κ. This project's own measurement puts the seed pairs' half-life at roughly a week, which is why the model rescans for new pairs every 20 days but estimates hedge ratios and means over a much longer 350–450 day window: long enough

for a stable β, reactive enough relative to the reversion speed.

$$t_{1/2} = \frac{\ln 2}{\kappa}$$

reversion half-life



Figure A.1. a simulated OU spread calibrated to a ~1-week half-life. The band shading marks ±1 and ±2 stationary standard deviations.

Once a pair is confirmed, the model trades a z-score of the current spread against its own rolling mean and EWMA-smoothed volatility, taking a full-size position whenever |z| clears a band and flattening otherwise (bang-bang sizing). Because both legs move together with the shared factor, the position is close to market-neutral within the pair; β is precisely the hedge ratio cancelling that shared component, leaving only the idiosyncratic spread to bet on.

KEY IDEA

A pair that cointegrated over its full history can still stop cointegrating recently. The recent-window kill switch (KEEPR) is not decoration: it guards against betting on convergence from a relationship that has already broken.

B. The Two-Factor Lead-Lag Matrix

Present in every model. Beyond pairs, there may be a broader but individually weak linear structure: today's move in name i can carry information about tomorrow's move in name j. The code estimates this directly as a dense 50×50 one-lag cross-correlation matrix L, which is a restricted Vector Autoregression, VAR(1):

$$x_{i,\,t+1} \;=\; \sum_{j} L_{j,\,i}\, x_{j,\,t} \;+\; \varepsilon_{i,\,t+1}$$

dense one-lag cross-forecast

Before L is fit, returns are residualized against ALGO (the market factor, correlation ≈0.99 with the equal-weight basket) and then against the first principal component of what remains (NFAC=2). This mirrors Arbitrage Pricing Theory (Ross, 1976) and Fama–French-style factor residualization: stripping the known market factor and the largest remaining latent factor (found via PCA, the eigenvector of the residual covariance matrix with the largest eigenvalue) isolates idiosyncratic, plausibly more local structure rather than rediscovering “the whole market moved together yesterday.”

Any single entry L[i,j] is a noisy correlation over 50–450 observations and, alone, is close to statistically insignificant. But the aggregate forecast for name j sums 49 such terms. This is a breadth argument, formalized in active-management theory as Grinold's Fundamental Law:

$$IR \;\approx\; IC\sqrt{BR}$$

Grinold's Fundamental Law of Active Management

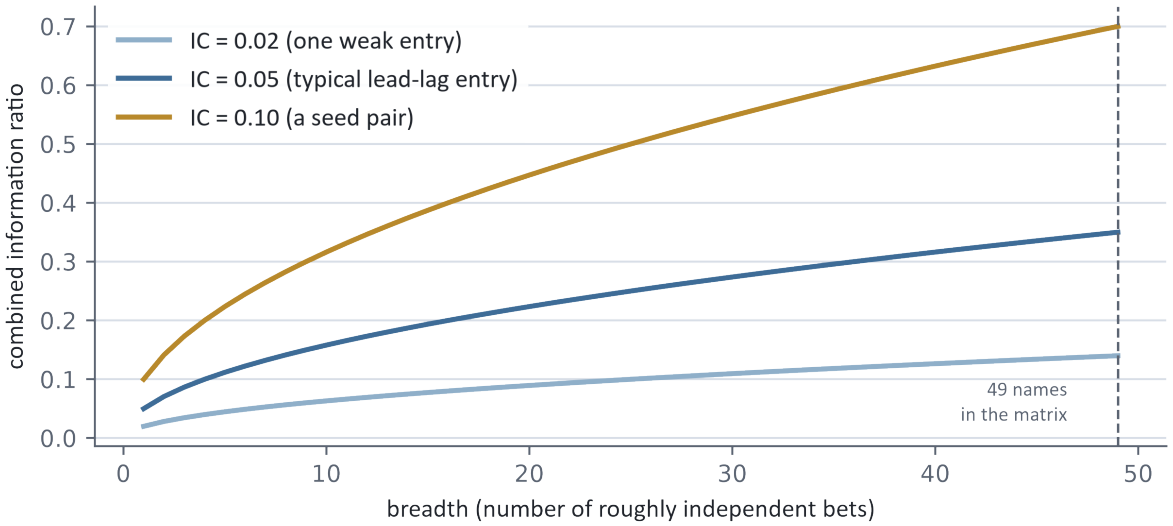


Figure B.1. combined information ratio as a function of breadth, for a few illustrative per-signal information coefficients. Many weak, roughly independent bets can out-perform a few clean ones.

This is exactly why the dense, unfiltered matrix beat every sparser alternative this project tried (top-K sparsification, shrinkage, ridge): each of those reduces breadth without a compensating gain in per-entry quality. The one reformulation that DID beat dense linear lead-lag in this project's own testing, a LASSO-selected sparse matrix, still failed the two-window bar once tested honestly on the other window, and was not shipped.

The “recency blend” refinement (SigmaB10 onward: a 90% expanding-history / 10% trailing-450-day blend) is a bias-variance tradeoff: the full-history matrix has lower variance but can be stale if lead-lag relationships drift; blending in a small amount of a recent-only estimate lets the signal track drift without giving up most of the statistical power of the long history.

KEY IDEA

Breadth is not a consolation prize for weak signals, it is the mechanism. A matrix of 2,450 individually-marginal correlations can carry more real information, in aggregate, than a handful of clean ones, provided the average edge stays positive and the bets are not all the same bet in disguise.

C. The Gray / Survivor Tier

Present in Sigma8 onward. Beyond the 7 hand-verified seed pairs, a much larger population of candidates passes a looser cointegration gate (full-history $t < -3.4$, recent $t < -2.8$, versus a stricter gate for the “strict survivor” pairs). With 50 names there are $50 \cdot 49/2 = 1{,}225$ possible pairs, and under a pure-noise null a meaningful fraction will pass a loose t-stat screen by chance alone: the classic multiple-comparisons problem, the same phenomenon behind “data snooping” risk in any large factor search.

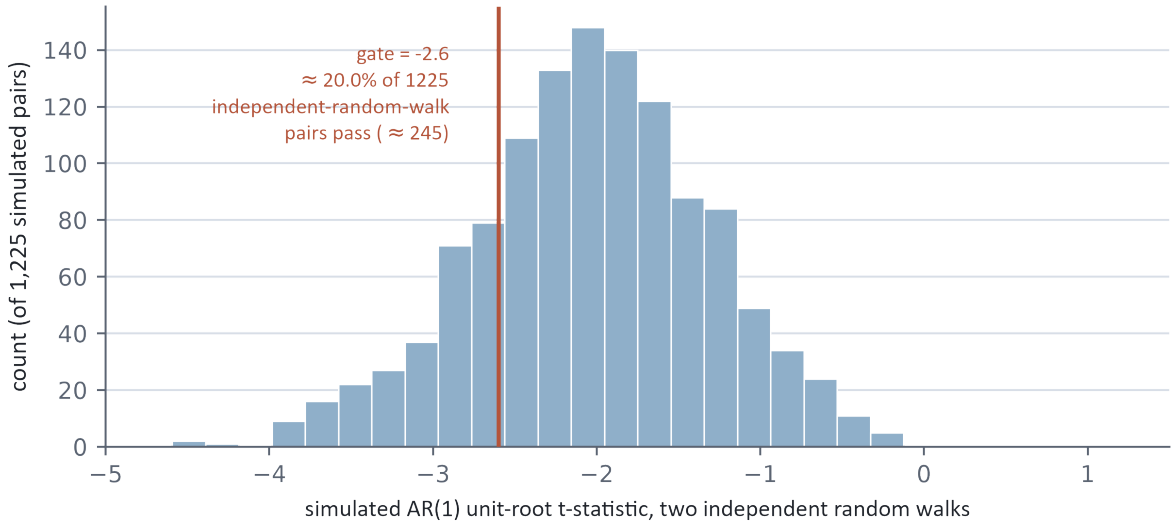


Figure C.1. simulated null distribution of pairwise AR(1) t-statistics from 1,225 pairs of pure random-walk noise. Even with no real structure at all, a non-trivial share clears a loose gate by chance; this is why the gate alone cannot be the only safeguard.

The mitigation is not to raise the bar until only “true” pairs remain (this project's own capacity sweep found doing so is strictly worse), but to trade the whole tier at half weight (`GW=0.5`) alongside the full-weight measured core. This is the breadth argument from Section B applied to portfolio construction: a large, diversified set of small, individually-noisy but on-average-positive bets can beat a small set of “cleaner” bets, and down-weighting (rather than fully trusting) the tier is a simple way to size for its lower average edge and higher false-positive rate.

KEY IDEA

With 1,225 candidate pairs, some will look cointegrated by pure chance. The fix used here is not a stricter filter (tried, and it hurt), it is smaller position size on the whole uncertain tier.

D. The Birth-Catcher

Present in Sigma8 onward, gate loosened from -4.0 to -3.5 starting with Sigma10FB. This detector looks for pairs cointegrated on a recent 150-day window but NOT (or only weakly) over the full history: it is explicitly hunting for relationships that did not exist before and have since appeared.

This matters here specifically because the project's own measurement of the price generator found the pair network rewires substantially over time. Between one 250-day window and the next, roughly 73% of edges turn over even though total pair count stays roughly stable (44 → 48). The tradeable population is not a fixed, discover-once set; it behaves like a non-stationary, regime-switching graph where old edges die and new ones are born.

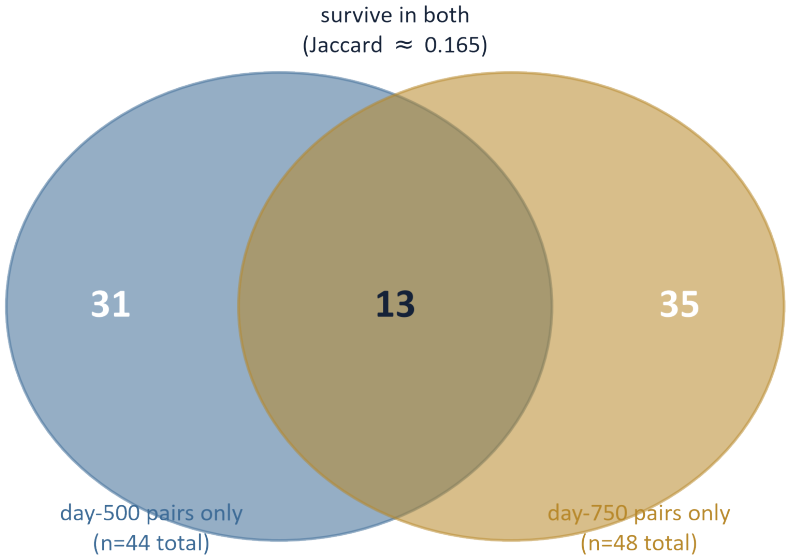


Figure D.1. the measured overlap between the pair set found at day 500 and the pair set found at day 750. Total count is roughly stable; identity is not (Jaccard similarity ≈ 0.165).

A static “find pairs once and trade them forever” strategy would progressively lose coverage as its discovered edges decay. The birth-catcher is a structural-break detector aimed squarely at this, adapting to newly-formed relationships without waiting on a full, slower historical rescan. Loosening its gate from -4.0 to -3.5 is a straightforward capacity increase: this project's own neighbourhood sweep (from -4.0 down to -3.2 in 0.1 steps) found a broad plateau roughly from -3.7 to -3.3, the signature of a real, robust effect rather than a lucky single value, and -3.5 sits in the middle of it.

KEY IDEA

The cointegration network here is not static. A detector tuned only to find pairs once, on day one, will silently lose coverage as the underlying relationships rewire; the birth-catcher exists to keep finding new ones.

E. Tail-Regime Diagnostics and Hysteresis

Present in SigmaB10 onward. The default assumption for both pair spreads and the lead-lag signal is that the relationship between “how far the signal is from zero” and “what happens next” is linear across the whole range. The diagnostic tests this directly, fitting two separate regression slopes: one using only central observations ($|z| < 1.5$) and one using only tail observations ($|z| \geq 2.0$), each against the realized next-period response.

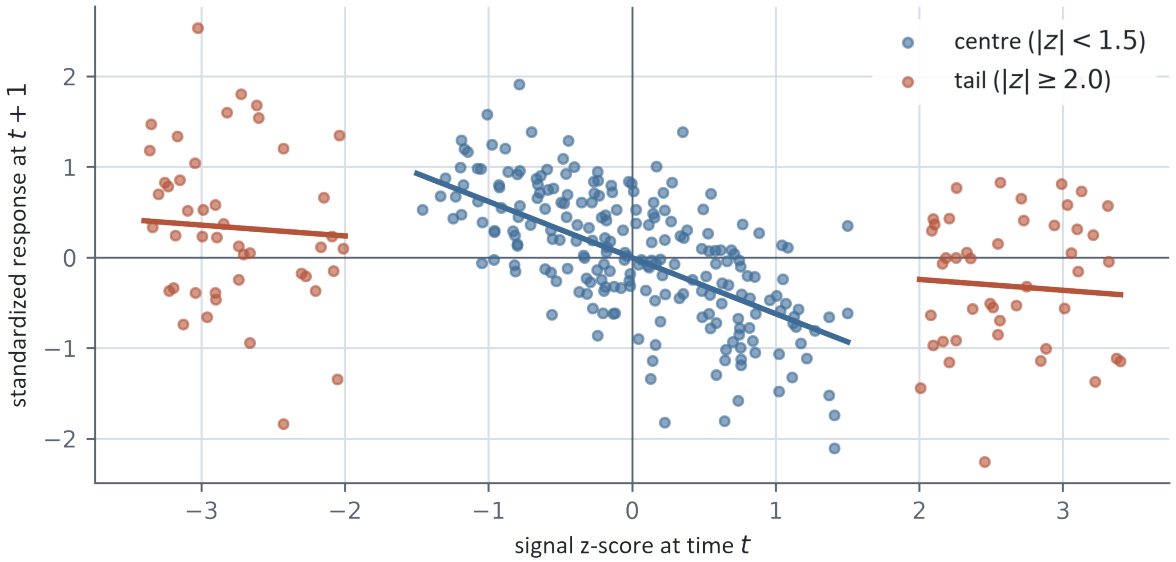


Figure E.1. a schematic illustration of the diagnostic's own logic: the centre reverts on a steep slope, the tail reverts on a much flatter one. When this gap is statistically reliable, the model treats tail readings differently.

If the tail slope is reliably weaker than the centre slope, the diagnostic concludes extreme readings do not revert the way moderate ones do. This connects to a few ideas at once: the empirical finding that market mean reversion is often regime-dependent (moderate deviations revert, very large ones can instead mark a structural break or the start of a trend); the general phenomenon of fat tails (leptokurtosis), where a linear, constant-speed OU model breaks down several standard deviations out; and, mechanically, a hysteresis rule.

Once tail-weakness is flagged, a position gains memory: it opens once |z| clears an entry band, then it is HELD, not flipped flat, unless the magnitude actually collapses back toward zero (an exit threshold) or blows out to a hard stop, at which point it is cut regardless of direction. This is the same dead-band, Schmitt-trigger pattern used in control theory to stop a system chattering on noise near a boundary, applied here as a stop-loss discipline: don't fight a move that has already invalidated the reversion assumption, and don't get shaken out of a good position by noise near the entry line either.

KEY IDEA

Not every extreme reading is a better bet than a moderate one. When the data show the tail reverting more weakly than the centre, the right response is a stop-loss-like hold/cut rule, not a bigger bet on the same linear model.

Part Two

The Exposure-Control Mechanics

ALGO (instrument 0) carries a 10× larger position limit ($100k vs $10k) and a 5× lower commission rate (0.02% vs 0.1%), so whatever occupies this slot has an outsized effect on total book performance. Every model computes one shared quantity, the dollar amount of ALGO that would exactly cancel the rest of the book's incidental exposure to the market factor:

$$h_t = -\sum_i w_{i,t} P_{i,t} \beta_i$$

beta-hedge construction

where β_i is each name's own regression coefficient on ALGO's returns. The five mechanics below differ only in what they do with this single computed quantity, or whether they use it at all.

1. Pure Beta Hedge (Sigma7p, Sigma8)

The ALGO position is set to exactly cancel the book's incidental exposure, full stop. This is the direct implementation of market-neutral construction, going back to the CAPM decomposition of any asset's return into a systematic and an idiosyncratic component:

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t}$$

CAPM return decomposition

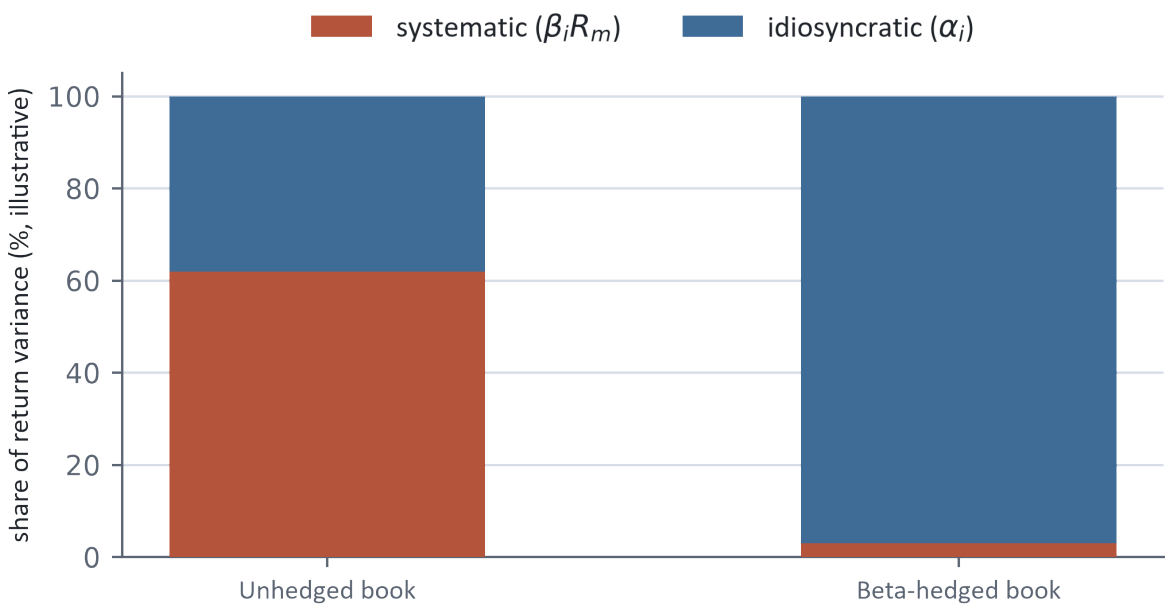


Figure 1.1. illustrative return-variance split before and after hedging. Cancelling the systematic component removes uncompensated variance without touching the idiosyncratic edge.

If the stat-arb/lead-lag book ends up net long or short the market factor by the accident of which names it happens to be positioned in, that exposure is not something the strategy has any forecasting edge on; it is uncompensated risk that adds variance without adding expected return. Cancelling it isolates pure alpha: keep the return from the measured edges, discard the return (and, more importantly, the extra variance) that rides along purely from position construction.

KEY IDEA

In Sigma7p and Sigma8 the ALGO slot is doing risk management, not forecasting: beta-neutral, no directional view.

2. Directional Reversal Bet `(SigmaB10, Sigma16, half of Sigma17)`

The hedge is switched off and ALGO becomes a forecastable instrument in its own right: bet that whatever ALGO has just done over 5 days will partially reverse, at the full $100k cap. This is the short-term reversal effect documented in the literature (Lehmann 1990, Jegadeesh 1990): at short, weekly-to-monthly horizons, returns tend to show negative serial correlation, commonly attributed to short-term overreaction, market-making inventory effects, or bid-ask bounce. It is the mirror image of momentum at 3–12 month horizons (Jegadeesh & Titman, 1993); reversal and momentum are different phenomena that dominate at different horizons, not contradictory claims.

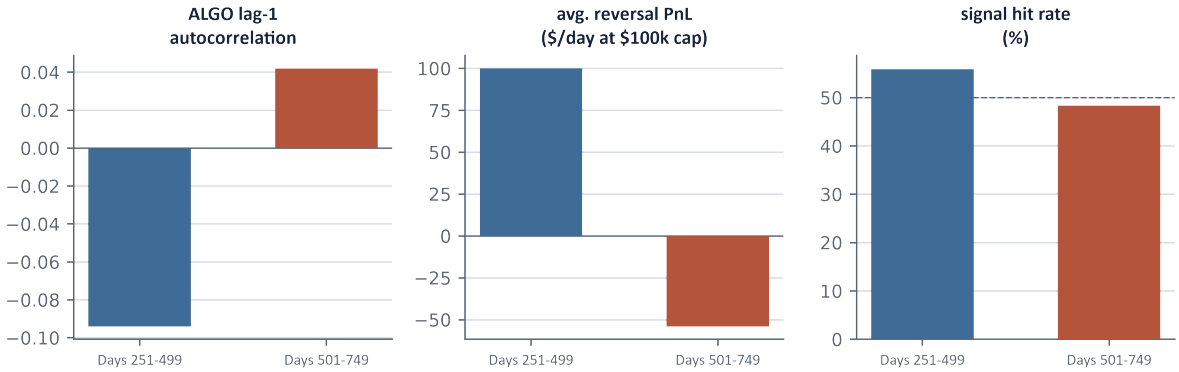


Figure 2.1. measured, not assumed. ALGO's own lag-1 autocorrelation, the reversal signal's average daily PnL, and its hit rate, on two different windows of the same generator.

The critical caveat, directly measured rather than assumed: on one window ALGO's lag-1 autocorrelation was -0.094 (genuinely mean-reverting, and the reversal signal was profitable at every horizon 1–15 days tested); on the next window it was +0.042 (the sign flipped: ALGO was now trending, and the same signal lost money at every horizon tested). Short-term reversal is an empirical regularity of a particular regime, not a physical law, and a strategy built on it is making an implicit, time-varying bet on which regime currently holds, whether or not that is stated explicitly.

KEY IDEA

The same 5-day-reversal signal earned roughly +$100/day on one window and roughly -$54/day on the very next one. The mechanism did not change; the regime did.

3. Anti-Hedge / Leverage `(Sigma10X)`

Instead of cancelling the book's incidental exposure, this mechanic amplifies it 1.5× in the same direction (the hedge weight is set to -1.5, which flips the sign of the computed hedge and scales it up). Superficially this looks like ordinary leverage, a normally value-additive tool: scale up a position with positive expected value to accept more variance for more expected return, the logic behind risk-parity and leveraged-beta strategies. Ordinary leverage on a genuine risk factor is defensible because that factor's premium is believed to be persistently positive over time.

But the book's “incidental exposure” here is not a risk factor with a persistent premium, it is simply whatever residual market-tilt falls out of running a market-neutral book on a given day, a byproduct of position construction with no design intent behind its sign. This project measured its correlation with ALGO's next-day return at -0.071 on one window and +0.106 on the other (the sign flips), and its own rolling 50-day block PnL:

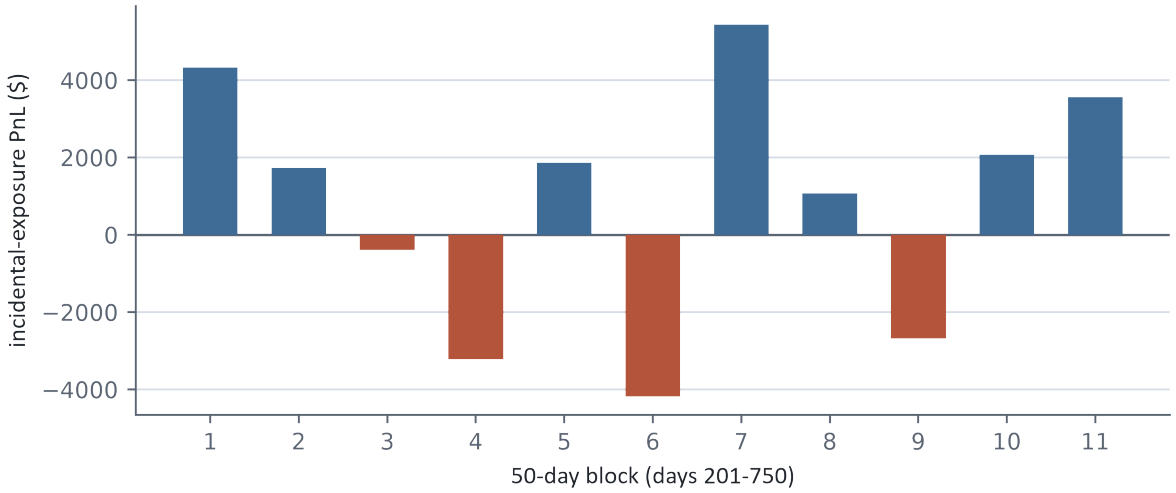


Figure 3.1. the incidental exposure term's actual 50-day block PnL, days 201-750. Consecutive same-sign blocks occurred only 4 of 10 transitions (below the ~50% a coin flip would give), and lag-1 autocorrelation of block PnL measured at -0.30 (mildly anti-persistent, i.e. it tends to alternate).

This is the statistical signature of noise, not a risk premium: a genuine risk factor's exposure should have a reasonably stable sign and a reasonably persistent premium; something with a coin-flip sign and no persistence has an expected value of zero, and leverage on it only multiplies variance for no expected-return benefit.

$$E[kX] = k\,E[X] \qquad \mathrm{Var}(kX) = k^2\,\mathrm{Var}(X)$$

leverage scales expectation and variance differently

This is the same warning underlying the Kelly criterion and any variance-aware sizing framework: confirm genuine positive expectancy (sign persistence, cross-window stability, exactly as measured here) BEFORE the leverage decision, not after a single lucky backtest window. In this project's own numbers, this mechanic scored 548 on one window and 806 on the other, a two-window average barely different from not touching the ALGO slot at all.

KEY IDEA

Leverage is only free upside on something with genuine positive expected value. Levering a zero-mean, sign-flipping residual just makes the variance bigger for the same expected return: zero.

4. Flat / Untraded `(Sigma10FB)`

The simplest of the five, and arguably the most useful for interpretation even though it is not really a trading strategy: both the reversal weight and the hedge weight are set to zero, so the ALGO slot is fully flat, always. This is a controlled ablation. By fully disabling the slot, this configuration isolates exactly what the rest of the

book (pairs, gray tier, birth-catcher, lead-lag) is worth on its own, with no confound from any ALGO-slot decision.

This is standard scientific method applied to strategy research: to measure what a component contributes, hold everything else fixed and remove only that component, then compare. Every other exposure mechanic in this document is best read relative to this flat baseline, since that is the cleanest available measure of what the ALGO slot itself adds or subtracts.

KEY IDEA

Sigma10FB is not really a strategy, it is the control group. Compare every other ALGO mechanic's score against it, not just against each other.

5. Relative Reversal `(half of Sigma17, weight 0.5)`

Where mechanic 2 fades ALGO's own move in isolation, this fades ALGO's move relative to the equal-weight basket of everything else: not “did ALGO go up,” but “did ALGO go up MORE than the rest of the market did,” over a 6-day horizon, blended 50/50 with the plain reversal signal:

$$r_t = -\left([\log P_{0,t} - \log P_{0,t-h}] - \sum_{i \neq 0} \bar{w}_i [\log P_{i,t} - \log P_{i,t-h}] \right)$$

relative-reversal trigger, horizon h

Structurally this is a relative-value or spread trade, the same family as the seed pairs in Section A, just with one leg being a single name (ALGO) and the other an equal-weighted basket of 49 names, sometimes called dispersion or relative-strength trading when applied to an index against its constituents.

The genuinely important nuance, again investigated rather than assumed: a correctly-lagged decomposition of this signal's realized PnL found the actual relative-value “spread” component contributed close to zero, sometimes slightly negative. Almost all of the measured edge came from the BASKET's own next-day return, using ALGO's relative underperformance merely as a trigger.

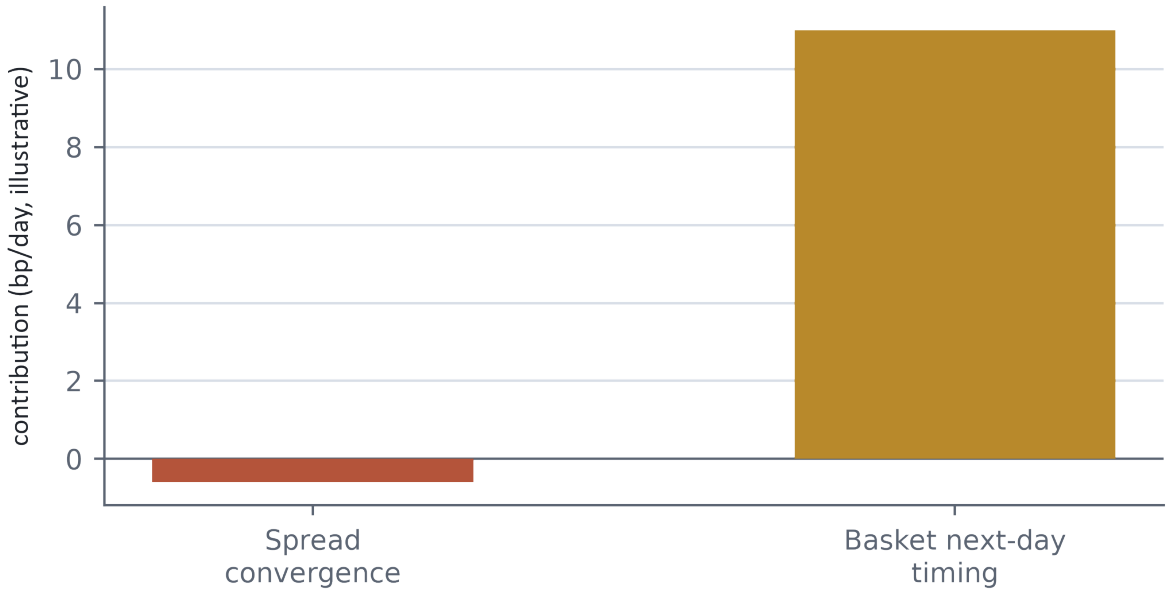


Figure 5.1. decomposition of the measured edge. Almost none of it is spread convergence; almost all of it is timing the basket's own next-day return.

This is not really a relative-value arbitrage trade, it is a market-factor-timing signal that happens to use a relative-value-shaped construction to generate its trigger. That distinction matters for how much to trust it going forward: it is the same broad category as mechanic 3, which was shown to be a zero-mean coinflip, and this one has not been proven immune to the same risk, it has simply not (yet, as of the last two available windows) shown the same sign-flipping instability. This project's own notes flag it as a carried-forward risk, not a settled question.

KEY IDEA

Naming a signal “relative value” does not make it one. Decomposing WHERE the PnL actually comes from revealed this is a market-timing bet wearing a relative-value costume.

Cross-Reference: Models to Mechanics

Each named model is a specific combination of Part One's building blocks, plus exactly one (or, for Sigma17, a blend of two) of Part Two's exposure mechanics. Tinted cells mark a component switched on; blank cells mark it switched off.

Model	Seed Pairs	Lead-Lag	Gray Tier	Birth Gate	Tail Hysteresis	ALGO Slot Mechanic
Sigma7p				off		Pure beta hedge
Sigma8				-4.0		Pure beta hedge
SigmaB10				-4.0		Directional reversal (5d)
Sigma10X				-4.0		Anti-hedge, 1.5x leverage
Sigma10FB				-3.5		Flat / untraded
Sigma16				-3.5		Directional reversal (5d)
Sigma17				-3.5		Reversal + rel. reversal, 50/50

Reading a model's entry in Model Evaluation Summary.txt alongside this table and the sections above should show not just WHAT each model does differently, but WHY that difference is expected, or in several cases empirically shown NOT, to matter.

Appendix: The Scoring Function

Every number in Model Performance Evaluation.txt is this function, applied to a 250-day window of daily PnL:

$$SR = \sqrt{250}\,\frac{\mu}{\sigma} \qquad \text{Score} = \mu \cdot \frac{SR^2}{SR^2 + 1}$$

annualized Sharpe and competition score

SR is the annualized Sharpe ratio of daily PnL. The score multiplies mean PnL by $SR^2/(SR^2+1)$, a factor that approaches 1 as Sharpe gets large and shrinks toward 0 as Sharpe falls, so raw mean PnL is heavily discounted unless it is earned with low variance. This single design choice explains a recurring pattern across this document: several mechanics above (breadth in the lead-lag matrix, the gray tier's diversification, half-weighting rather than excluding uncertain bets) are all, in one way or another, trading a little raw expected return for a large reduction in variance, because that trade is exactly what this scoring function rewards.

Companion documents: Model Evaluation Summary.txt (per-model architecture, exposure, and performance), Model Performance Evaluation.txt (scores only), and Model List/ (every model file this project has produced or received, including the Zigma externally-authored variants).